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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
21CSE583T – NATURAL LANGUAGE PROCESSING AND ITS
APPLICATIONS
CYCLE TEST – I

Program Offered: M. Tech
Max. Marks: 40

Year / Semester: I / I
Duration: 1.5 Hrs
Date of Exam:

Part - A (2 * 20 = 40 Marks)

Answer any two Questions

Model Question Paper

1)	You are working as a Data Scientist for a social media sentiment analysis project. Your company wants to analyze customer feedback from various social media platforms, which include slang, abbreviations, hashtags, emojis, and inconsistent capitalization. a) Explain the role of text preprocessing in NLP and why it is crucial for sentiment analysis. (10 marks) b) Discuss the significance of tokenization in NLP and compare different tokenization methods that would be effective for processing social media text. (10 marks)
2)	You are working as an NLP Engineer for an academic research paper recommendation system. The system aims to analyze research abstracts and suggest relevant papers to users based on their past readings. a) Explain the importance of text vectorization in NLP and its role in this research paper recommendation system. (10 marks) b) Compare three common text vectorization techniques (Bag of Words, TF-IDF, and Word Embeddings) with examples. (10 marks)

3)	You are working as an NLP Engineer for a book review platform. The company wants to automatically group book reviews into themes such as Mystery, Romance, Science Fiction, and Non-Fiction without manually labeling them. Your task is to identify hidden topics within the reviews. a) Describe the LDA (Latent Dirichlet Allocation) technique with examples and explain how it helps in topic discovery. (10 marks) b) What is a Word Cloud, and how can it assist in analyzing book reviews? (10 marks)
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1. You are working as a Data Scientist for a social media sentiment analysis project. Your company wants to analyze customer feedback from various social media platforms, which include slang, abbreviations, hashtags, emojis, and inconsistent capitalization.

- a) Explain the role of text preprocessing in NLP and why it is crucial for sentiment analysis. (10 marks)

Discuss the significance of tokenization in NLP and compare different tokenization methods that would be effective for processing social media text. (10 marks)

Answer:

a) Role of Text Preprocessing in NLP and Its Importance for Sentiment Analysis (10 marks)

Role of Text Preprocessing in NLP:

Text preprocessing is a critical step in Natural Language Processing (NLP) that involves transforming raw text data into a clean, structured format suitable for analysis. It includes tasks such as:

1. **Noise Removal:** Eliminating irrelevant characters, symbols, or HTML tags.
2. **Normalization:** Converting text to a consistent format (e.g., lowercasing, expanding contractions, handling slang/abbreviations).
3. **Tokenization:** Breaking text into smaller units like words or sentences.
4. **Stopword Removal:** Removing common words (e.g., "the," "is") that add little meaning.
5. **Handling Emojis and Hashtags:** Converting emojis and hashtags into meaningful text representations.
6. **Lemmatization/Stemming:** Reducing words to their base or root form.

Why It Is Crucial for Sentiment Analysis:

1. **Improves Model Accuracy:** Preprocessing ensures that the input data is consistent and free of noise, which helps machine learning models focus on meaningful patterns.
2. **Handles Variability:** Social media text often contains slang, abbreviations, and inconsistent capitalization, which preprocessing can standardize.
3. **Enables Feature Extraction:** Cleaned and normalized text makes it easier to extract features like sentiment-bearing words or phrases.
4. **Handles Emojis and Hashtags:** Emojis and hashtags often carry sentiment (e.g., 😊 or #happy), and preprocessing ensures they are interpreted correctly.
5. **Reduces Dimensionality:** Techniques like stopword removal and stemming reduce the complexity of the data, making it easier to process.

Without preprocessing, the sentiment analysis model might misinterpret or miss important sentiment cues, leading to inaccurate results.

b) Significance of Tokenization in NLP and Comparison of Tokenization Methods for Social Media Text (10 marks)

Significance of Tokenization in NLP:

Tokenization is the process of splitting text into smaller units, such as words, phrases, or sentences. It is a foundational step in NLP because:

1. **Enables Text Analysis:** Tokenization breaks down text into manageable units for further processing, such as sentiment analysis or part-of-speech tagging.
2. **Facilitates Feature Extraction:** Tokens serve as the basis for creating features (e.g., word frequency, n-grams) for machine learning models.
3. **Handles Contextual Meaning:** Proper tokenization ensures that words, emojis, and hashtags are treated as individual units, preserving their meaning.

Comparison of Tokenization Methods for Social Media Text:

1. Whitespace Tokenization:

- **How It Works:** Splits text based on spaces.
- **Pros:** Simple and fast.
- **Cons:** Fails to handle punctuation, emojis, or hashtags effectively (e.g., "😊" or "#happy" might not be tokenized correctly).
- **Suitability for Social Media:** Not ideal due to the complexity of social media text.

2. Word Tokenization (e.g., NLTK, spaCy):

- **How It Works:** Splits text into words, handling punctuation and contractions.
- **Pros:** More accurate than whitespace tokenization; works well for standard text.
- **Cons:** May struggle with slang, abbreviations, or emojis.
- **Suitability for Social Media:** Better than whitespace but may require additional preprocessing for social media-specific features.

3. Tweet Tokenizer (e.g., NLTK's TweetTokenizer):

- **How It Works:** Specifically designed for social media text; handles hashtags, emojis, and mentions.
- **Pros:** Effective for social media data; preserves emojis and hashtags as tokens.
- **Cons:** May still require additional normalization for slang or abbreviations.
- **Suitability for Social Media:** Highly suitable due to its focus on social media text.

4. Subword Tokenization (e.g., Byte Pair Encoding - BPE):

- **How It Works:** Splits words into subword units (e.g., "unhappiness" → "un", "happiness").
- **Pros:** Handles out-of-vocabulary words and rare terms effectively.
- **Cons:** Can be computationally expensive and less interpretable.
- **Suitability for Social Media:** Effective for handling slang, abbreviations, and misspellings.

5. Emoji and Hashtag-Aware Tokenization:

- **How It Works:** Treats emojis and hashtags as separate tokens while preserving their meaning.
- **Pros:** Captures sentiment and context from emojis and hashtags.
- **Cons:** Requires custom implementation or specialized libraries.
- **Suitability for Social Media:** Highly effective for sentiment analysis on social media.

Conclusion:

For social media sentiment analysis, **Tweet Tokenizer** and **Emoji/Hashtag-Aware Tokenization** are the most effective methods. They handle the unique characteristics of social media text, such as emojis, hashtags, and slang, ensuring that sentiment-bearing elements are preserved and analyzed accurately.

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| 2 | <p>You are working as an NLP Engineer for an academic research paper recommendation system. The system aims to analyze research abstracts and suggest relevant papers to users based on their past readings.</p> <p>c) Explain the importance of text vectorization in NLP and its role in this research paper recommendation system. (10 marks)</p> |
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| | d) Compare three common text vectorization techniques (Bag of Words, TF-IDF, and Word Embeddings) with examples. (10 marks) |
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a) Importance of Text Vectorization in NLP and Its Role in the Research Paper Recommendation System

Text Vectorization is a fundamental step in Natural Language Processing (NLP) that converts textual data into numerical representations, enabling machine learning models to process and analyze text. In the context of a research paper recommendation system, text vectorization plays a critical role in the following ways:

1. **Feature Representation:** Research paper abstracts are rich in textual information, but machine learning models cannot directly process raw text. Vectorization transforms the text into numerical features (e.g., vectors) that capture the semantic and syntactic properties of the text, making it usable for algorithms.
2. **Similarity Measurement:** The recommendation system relies on finding similarities between papers based on their abstracts. Vectorization allows the system to compute similarity scores (e.g., cosine similarity) between papers by comparing their vector representations.
3. **Dimensionality Reduction:** Abstracts can be lengthy and complex, but vectorization techniques like TF-IDF or word embeddings reduce the dimensionality of the data while preserving meaningful information, improving computational efficiency.
4. **Personalization:** By vectorizing user reading histories and paper abstracts, the system can match user preferences with relevant papers, enabling personalized recommendations.
5. **Model Compatibility:** Most recommendation algorithms (e.g., collaborative filtering, content-based filtering, or deep learning models) require numerical input. Vectorization bridges the gap between textual data and these algorithms.

In summary, text vectorization is essential for transforming unstructured text into structured numerical data, enabling the system to analyze, compare, and recommend research papers effectively.

b) Comparison of Three Common Text Vectorization Techniques

Here are three widely used text vectorization techniques, along with their strengths, weaknesses, and examples:

1. Bag of Words (BoW)

- **Description:** BoW represents text as a vector of word frequencies. It creates a vocabulary of unique words from the corpus and counts how often each word appears in a document.
- **Example:**
 - Document 1: "The cat sat on the mat."
 - Document 2: "The dog sat on the log."
 - Vocabulary: {"the", "cat", "sat", "on", "mat", "dog", "log"}
 - BoW Vectors:
 - Document 1: [2, 1, 1, 1, 1, 0, 0]
 - Document 2: [2, 0, 1, 1, 0, 1, 1]

- **Advantages:**
 - Simple and easy to implement.
 - Works well for small datasets.
- **Disadvantages:**
 - Ignores word order and context.
 - Sparse and high-dimensional vectors for large vocabularies.
 - Fails to capture semantic relationships between words.

2. Term Frequency-Inverse Document Frequency (TF-IDF)

- **Description:** TF-IDF improves upon BoW by weighting words based on their importance in a document relative to the entire corpus. It combines term frequency (TF) and inverse document frequency (IDF).
- **Example:**
 - Document 1: "The cat sat on the mat."
 - Document 2: "The dog sat on the log."
 - TF-IDF Vectors (example weights):
 - Document 1: [0.2, 0.8, 0.3, 0.1, 0.7, 0.0, 0.0]
 - Document 2: [0.2, 0.0, 0.3, 0.1, 0.0, 0.8, 0.7]

- **Advantages:**
 - Reduces the importance of common words (e.g., "the") and highlights unique, meaningful words.
 - Better at capturing document-specific relevance.
- **Disadvantages:**
 - Still ignores word order and context.
 - Sparse vectors for large vocabularies.

3. Word Embeddings (e.g., Word2Vec, GloVe, FastText)

- **Description:** Word embeddings represent words as dense, low-dimensional vectors in a continuous vector space, capturing semantic relationships between words.
- **Example:**
 - Word2Vec embeddings for "king," "queen," "man," and "woman" might show relationships like: $\text{king} - \text{man} + \text{woman} \approx \text{queen}$.
 - Document representation can be obtained by averaging word embeddings or using advanced methods like Doc2Vec.
- **Advantages:**
 - Captures semantic and syntactic relationships between words.
 - Dense vectors reduce dimensionality and sparsity.
 - Can generalize to unseen words (e.g., in FastText).

- **Disadvantages:**

- Requires large datasets and computational resources for training.
- Pre-trained embeddings may not be domain-specific (e.g., academic papers).

Comparison Summary:

Technique	Strengths	Weaknesses
Bag of Words	Simple, easy to implement, works for small datasets.	Ignores context, sparse vectors, high-dimensional.
TF-IDF	Weights words by importance, reduces noise from common words.	Still sparse, ignores word order and context.
Word Embeddings	Captures semantic relationships, dense vectors, reduces dimensionality.	Requires large datasets, computationally expensive, may need domain adaptation.

Conclusion

For a research paper recommendation system, **word embeddings** are often the best choice due to their ability to capture semantic relationships and produce dense, meaningful representations. However, **TF-IDF** can be a good alternative for smaller datasets or when computational resources are limited. **Bag of Words** is less suitable for this task due to its inability to capture context and semantics. The choice of technique depends on the dataset size, computational resources, and the desired level of semantic understanding.

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| 3 | <p>You are working as an NLP Engineer for a book review platform. The company wants to automatically group book reviews into themes such as Mystery, Romance, Science Fiction, and Non-Fiction without manually labeling them. Your task is to identify hidden topics within the reviews.</p> <p>c) Describe the LDA (Latent Dirichlet Allocation) technique with examples and explain how it helps in topic discovery. (10 marks)</p> <p>d) What is a Word Cloud, and how can it assist in analyzing book reviews? (10 marks)</p> |
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a) Latent Dirichlet Allocation (LDA) Technique

Description:

Latent Dirichlet Allocation (LDA) is a generative probabilistic model used in natural language processing (NLP) to discover hidden topics within a collection of documents. It assumes that each document is a mixture of a small number of topics, and each topic is characterized by a distribution over words. LDA is an unsupervised learning technique, meaning it does not require labeled data to identify topics.

How LDA Works:

1. **Document-Topic Distribution:** LDA assumes that each document is a mixture of topics. For example, a book review might be 60% about "Mystery" and 40% about "Romance."
2. **Topic-Word Distribution:** Each topic is represented as a distribution over words. For instance, the "Mystery" topic might have high probabilities for words like "clue," "detective," and "suspense," while the "Romance" topic might have high probabilities for words like "love," "heart," and "passion."
3. **Generative Process:** LDA models the process by which documents are generated:
 - For each document, a distribution over topics is chosen.
 - For each word in the document, a topic is selected from the document's topic distribution.
 - A word is then chosen from the topic's word distribution.

Example:

Consider a set of book reviews. LDA might identify topics such as:

- **Mystery:** Words like "clue," "detective," "suspense," "murder," "investigation."
- **Romance:** Words like "love," "heart," "passion," "relationship," "kiss."
- **Science Fiction:** Words like "space," "future," "alien," "technology," "planet."
- **Non-Fiction:** Words like "fact," "history," "research," "data," "evidence."

How LDA Helps in Topic Discovery:

- **Unsupervised Learning:** LDA does not require pre-labeled data, making it suitable for discovering unknown themes in book reviews.
- **Topic Interpretation:** By examining the top words associated with each topic, you can interpret and label the topics (e.g., "Mystery," "Romance").
- **Document Classification:** LDA can assign topics to new reviews, helping in organizing and categorizing them automatically.
- **Dimensionality Reduction:** LDA reduces the high-dimensional word space to a lower-dimensional topic space, making it easier to analyze and visualize the data.

b) Word Cloud and Its Role in Analyzing Book Reviews

Description:

A Word Cloud is a visual representation of text data where words are displayed in different sizes based on their frequency or importance. The more frequently a word appears in the text, the larger and bolder it appears in the Word Cloud.

How Word Clouds Assist in Analyzing Book Reviews:

1. **Quick Overview:** Word Clouds provide a quick and intuitive way to grasp the most prominent terms in a collection of book reviews. For example, if "love," "mystery," and "adventure" appear frequently, it suggests that these themes are common in the reviews.
2. **Theme Identification:** By looking at the most prominent words, you can identify common themes or topics in the reviews. For instance, a Word Cloud dominated by words like "space," "alien," and "future" might indicate a strong presence of Science Fiction themes.
3. **Sentiment Analysis:** Word Clouds can also give hints about the sentiment of the reviews. Words like "amazing," "thrilling," and "heartwarming" suggest positive sentiment, while words like "disappointing," "boring," and "confusing" suggest negative sentiment.
4. **Comparative Analysis:** You can create Word Clouds for different sets of reviews (e.g., by genre, by author) to compare the dominant themes and sentiments across these sets.
5. **Data Cleaning Insights:** Word Clouds can reveal stop words or irrelevant terms that might need to be removed or stemmed in further text analysis.

Example:

Imagine a Word Cloud generated from a set of book reviews:

- **Large Words:** "love," "mystery," "adventure," "thrilling," "heartwarming."
- **Smaller Words:** "predictable," "slow," "confusing," "disappointing."

This Word Cloud suggests that the reviews are generally positive, with themes of love, mystery, and adventure being prominent. However, there are also some negative sentiments indicated by words like "predictable" and "disappointing."

Conclusion:

Both LDA and Word Clouds are powerful tools for analyzing book reviews. LDA helps in discovering hidden topics and organizing reviews into themes, while Word Clouds provide a quick visual summary of the most common words and themes, aiding in initial exploration and interpretation of the data. Together, they can significantly enhance the understanding and categorization of book reviews on a platform.

